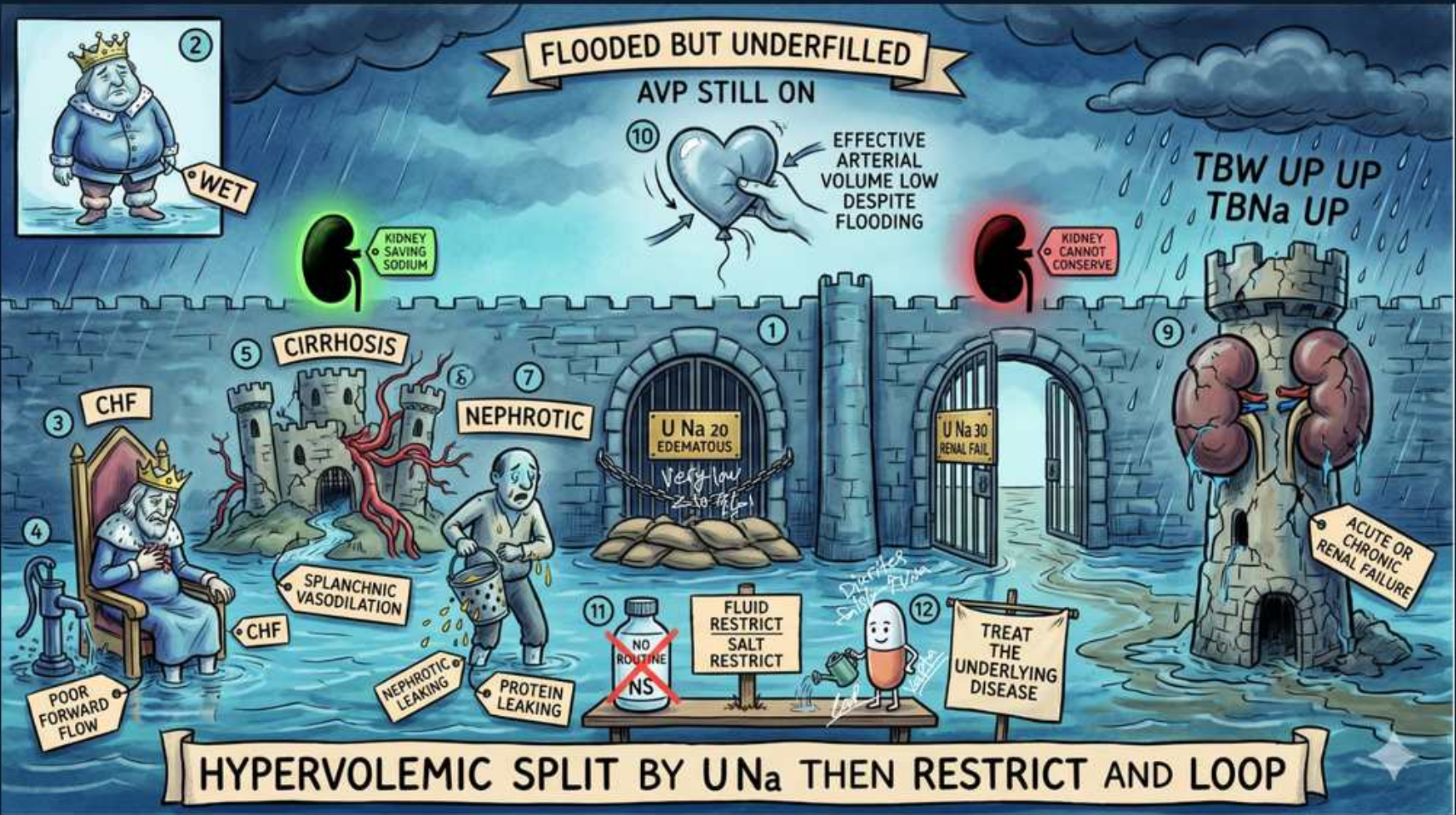


Hyponatremia — Hypervolemic (Edematous States)



1. Flooded but underfilled

WHAT YOU SEE

Banner: FLOODED BUT UNDERFILLED — AVP STILL ON

WHAT IT ENCODES

Edematous states (HF, cirrhosis, nephrotic) are total-body Na and water OVERLOADED yet have LOW effective arterial blood volume (the volume the baroreceptors sense). The kidney and carotid/aortic baroreceptors perceive underfilling, so RAAS and non-osmotic ADH stay ON, retaining free water and diluting serum Na. The result is dilutional hyponatremia in a clinically WET patient.

BOARD TRAP

WRONG: reasoning that an edematous, fluid-overloaded patient "has too much volume" for ADH to be active. The EFFECTIVE arterial volume is low despite total-body overload, which is exactly why ADH stays on — the discriminator is effective circulating volume, not total body water.

2. Split by UNa then restrict + loop

WHAT YOU SEE

Bottom banner: HYPERVOLEMIC SPLIT BY U Na THEN RESTRICT AND LOOP

WHAT IT ENCODES

Within hypervolemic hyponatremia, urine Na splits the causes: Una <20 in HF/cirrhosis/nephrotic (intact tubules avidly retaining Na under RAAS/ADH) versus Una >20–30 in acute/chronic renal failure (tubules cannot conserve Na). Core treatment for all: fluid AND salt restriction plus a LOOP diuretic, while treating the underlying disease.

BOARD TRAP

WRONG: giving isotonic or hypertonic saline to "correct the sodium" in an edematous patient. It adds Na and water to an already overloaded body and worsens edema/congestion — the answer is fluid + salt restriction plus loop diuresis (vaptans in select cases).

3. TBW up, TBNa up

WHAT YOU SEE

Text: TBW UP UP, TBNa UP UP at upper right

WHAT IT ENCODES

In hypervolemic hyponatremia BOTH total body water and total body sodium are increased, but WATER is gained proportionally more than Na — hence the dilutional fall in serum Na despite Na overload. Clinically this manifests as edema, ascites, pulmonary congestion, and weight gain.

BOARD TRAP

WRONG: assuming low serum Na means low total-body Na. Here total-body Na is actually HIGH; the hyponatremia is purely dilutional from even greater water excess. Treating with salt loading is exactly backwards.

4. Effective arterial volume low

WHAT YOU SEE

Broken heart: EFFECTIVE ARTERIAL VOLUME LOW DESPITE FLOODING

WHAT IT ENCODES

Low effective arterial blood volume — from poor forward cardiac output (HF), splanchnic arterial vasodilation (cirrhosis), or intravascular underfilling (nephrotic hypoalbuminemia) — is the shared trigger. Baroreceptors sense underfilling and activate RAAS, sympathetic tone, and non-osmotic ADH release, driving Na and water retention.

BOARD TRAP

WRONG: thinking the neurohormonal activation is inappropriate/idiopathic like SIADH. It is an APPROPRIATE response to perceived hypovolemia (low effective volume) — which is why the treatment targets the underlying low-output/underfilled state, not ADH alone.

5. CHF king

WHAT YOU SEE

Crowned figure labeled CHF, POOR FORWARD FLOW

WHAT IT ENCODES

Heart failure: reduced cardiac output lowers effective arterial volume, activating RAAS, sympathetics, and ADH → avid Na retention and free-water retention → dilutional hyponatremia. The degree of hyponatremia correlates with HF severity and is a marker of poor prognosis. Treat with fluid/salt restriction, loop diuretics, and guideline-directed HF therapy.

BOARD TRAP

WRONG: treating HF hyponatremia with saline or liberalized fluids. This worsens congestion. The Na improves as forward flow improves with diuresis and HF therapy; vaptans (tolvaptan) may be added for refractory symptomatic hyponatremia.

6. Cirrhosis castle

WHAT YOU SEE

Castle labeled CIRRHOSIS, SPLANCHNIC VASODILATION

WHAT IT ENCODES

Cirrhosis: portal hypertension drives splanchnic arterial vasodilation (NO-mediated), dropping effective arterial volume despite massive ascites. RAAS/ADH activation causes Na and water retention; hyponatremia is a marker of advanced disease and worse transplant outcomes. Treat with fluid/salt restriction; vaptans may help select cases.

BOARD TRAP

WRONG: aggressive saline or rapid correction in cirrhotic hyponatremia. These patients tolerate overcorrection poorly and are at HIGH risk for osmotic demyelination — correct slowly (≤6–8 mEq/L/24h) and prioritize fluid restriction and treating the portal hypertension/ascites.

7. Nephrotic

WHAT YOU SEE

Figure labeled NEPHROTIC, PROTEIN LEAKING, INTRAVASC LEAKING

WHAT IT ENCODES

Nephrotic syndrome: heavy proteinuria → hypoalbuminemia → reduced plasma oncotic pressure → intravascular fluid shifts to the interstitium → low effective arterial volume → ADH/RAAS activation with Na and water retention and dilutional hyponatremia. (Some nephrotic edema is also from primary renal Na retention.)

BOARD TRAP

WRONG: attributing nephrotic edema solely to "too much salt intake" and missing the underfill/ADH mechanism. Management is fluid/salt restriction plus loop diuretics and treating the underlying glomerular disease — not saline repletion.

8. Renal failure tower

WHAT YOU SEE

Tower labeled ACUTE OR CHRONIC RENAL FAILURE

WHAT IT ENCODES

Renal failure causes hypervolemic hyponatremia because the failing kidney cannot excrete the ingested water/Na load. Unlike HF/cirrhosis/nephrotic, the tubules CANNOT conserve Na, so urine Na is HIGH (>20–30). Management centers on water/salt restriction and, when needed, dialysis or diuretics if residual function exists.

BOARD TRAP

WRONG: expecting a low urine Na in renal-failure hyponatremia. The damaged tubules leak Na, so Una is >20–30 — the opposite of the avid Na retention seen in HF/cirrhosis/nephrotic. This urine-Na split is a classic exam discriminator.

9. Edematous gate UNa<20

WHAT YOU SEE

Left door: U Na 20 EDEMATOUS

WHAT IT ENCODES

In HF, cirrhosis, and nephrotic syndrome the tubules are intact and respond to RAAS/ADH by avidly reabsorbing sodium, so urine Na is LOW (<20 mEq/L). This is the same renal Na-conservation signature seen in true hypovolemia, reflecting the shared low effective arterial volume.

BOARD TRAP

WRONG: using a low urine Na (<20) to call a patient hypovolemic when they are clinically edematous. Edematous (low-effective-volume) states ALSO retain Na avidly — combine urine Na with the volume exam (edema/ascites/JVD) to place the patient correctly.

10. Renal failure gate UNa>30

WHAT YOU SEE

Right door: U Na 30 RENAL FAIL

WHAT IT ENCODES

In acute or chronic renal failure the kidney loses its concentrating and Na-conserving ability, so urine Na is HIGH (>20–30) even though the patient is volume-overloaded. This is how renal failure is separated from the avidly Na-retaining edematous states within the hypervolemic category.

BOARD TRAP

WRONG: interpreting a high urine Na as proof of SIADH or salt-wasting in an overloaded renal-failure patient. In renal failure the high Una reflects tubular inability to conserve, not appropriate water-handling — anchor on volume status and creatinine.

11. Kidney saving / cannot conserve

WHAT YOU SEE

Green kidney KIDNEY SAVING SODIUM, red kidney KIDNEY CANNOT CONSERVE

WHAT IT ENCODES

Two opposite renal behaviors define the split: in HF/cirrhosis/nephrotic the healthy kidney HOARDS sodium (green kidney, Una <20) under RAAS/ADH; in renal failure the diseased kidney CANNOT conserve sodium (red kidney, Una >20–30). Same hypervolemic dilutional hyponatremia, different urine Na.

BOARD TRAP

WRONG: applying one urine-Na cutoff to all hypervolemic hyponatremia. The cutoff direction depends on whether tubules are intact (retain, Una<20) or failing (leak, Una>30) — you must know which mechanism you're in.

12. Fluid + salt restrict

WHAT YOU SEE

Sign: FLUID RESTRICT SALT RESTRICT with NO Na bottle

WHAT IT ENCODES

The mainstay across all hypervolemic hyponatremia is restricting BOTH free water and dietary sodium to reverse the dilutional overload. Fluid restriction addresses the water excess (and ADH-driven retention); salt restriction addresses the total-body Na excess and edema. This is combined with loop diuretics and treating the underlying disease.

BOARD TRAP

WRONG: restricting fluid but allowing a normal/high salt diet (or vice versa). Both must be limited — leaving salt unrestricted perpetuates edema, while ignoring water keeps Na low. The combination, not one alone, is correct.

13. Loop diuretic

WHAT YOU SEE

Pill near TREAT THE UNDERLYING DISEASE sign

WHAT IT ENCODES

Loop diuretics (furosemide, etc.) are preferred in edematous hyponatremia: they excrete Na and produce relatively dilute urine, removing proportionally more free water and helping raise serum Na while decongesting. Combine with treating the underlying HF/cirrhosis/nephrotic. Vaptans (tolvaptan) are an option for refractory hypervolemic hyponatremia in HF/cirrhosis.

BOARD TRAP

WRONG: using a THIAZIDE (or normal saline) in hypervolemic hyponatremia. Thiazides impair urinary dilution and can WORSEN hyponatremia, and saline adds to the overload — the answer is a LOOP diuretic plus fluid/salt restriction.

14. Wet king reminder

WHAT YOU SEE

Inset crowned obese figure labeled WET, KIDNEY SAVING SODIUM

WHAT IT ENCODES

Hypervolemic hyponatremia is clinically WET — peripheral edema, ascites, pulmonary crackles, elevated JVD, and weight gain. This exam picture is the key anchor distinguishing it from euvoletic (normal exam, e.g., SIADH) and hypovolemic (dry, orthostatic) hyponatremia, and it drives the restrict-and-diuresis strategy.

BOARD TRAP

WRONG: confusing a wet, edematous hyponatremic patient with SIADH and prescribing only fluid restriction without diuresis, or worse, giving saline. The edema ("wet king") signals hypervolemia — treat with fluid + salt restriction AND a loop diuretic, plus the underlying disease.